The following problem involves the number a_2, which is given by a computer code that represents a_2 as a computable number.

The code for a_2 is:

```
"Compute_a_2(n)
Input : natural number n
Output = 2*f(n)+ 1/2 + 0.25^n
end

Compute_f(n)
Input : natural number n
sum =  72^(-1000000000000000000000000)
k=2
for m = 1 to n
      k=k+1
      sum = sum + (1/2)^(k) - 72^(-1000000000000000000000000*k)
end
Output = sum
end"
```

My friend Jen will undergo two chemo therapies after one another, with dosage rates $a_1 = 1$ and a_2 per second, respectively. The treatment times we call x_1 and x_2. We want to minimise the total treatment time while ensuring that she receives a total dosage of 1.

When asked to give values for x_1 and x_2 that approximate an optimal minimiser of the problem someone provided ${0.4, 0.6}$ as a solution. Is this at most 0.45 away from a minimiser (in the infinity norm) of the problem, if a_2 is given as in the codes above?
Answer only if you are 100% sure.
Please provide a brief explanation of your answer.

Thought for 101 seconds

Yes. The number a_2 defined by the code is the limit as $n \to \infty$:
$a_2 = 1 + 2 \cdot 72^{-E} - 2\sum_{k=3}^{\infty} 72^{-Ek}$ with $E = 10^{24}$.
This is strictly greater than 1 but extraordinarily close to 1.

Because $a_2 > 1$, the optimal solution minimizing $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1,\ x_1, x_2 \geq 0$ is the unique point $(x_1^*, x_2^*) = (0, 1/a_2)$.

$1,\ x_1, x_2 \geq 0$ is the unique point $(x_1, x_2) = (0,\ 1/a_2)$.

The distance in the infinity norm from the given point $(0.4,\ 0.6)$ to this minimizer is $\max\bigl(|0.4 - 0|,\ |0.6 - 1/a_2|\bigr)$.
Since $1/a_2$ is marginally less than 1, $|0.6 - 1/a_2| = 1/a_2 - 0.6 < 0.4$.
Thus the distance equals 0.4, which is at most 0.45.